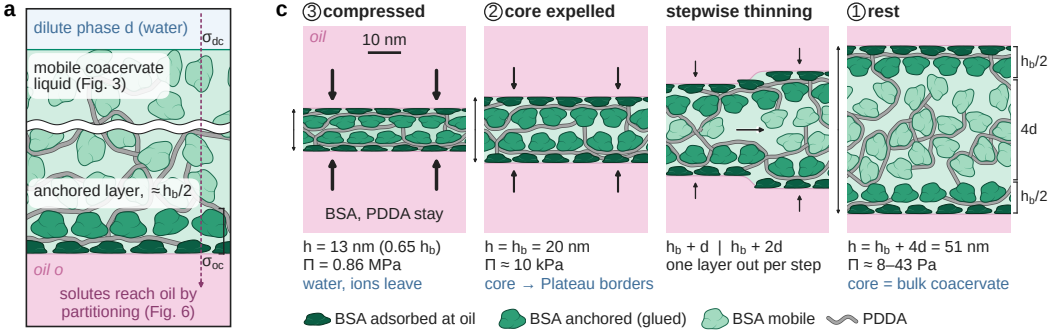
Fig. 1 molecular schematics: a, corona zoom; c, film states (Π(h) order)

Built 1:1. Zoom 1 nm = 0.6 mm; film states 1 nm = 0.44 mm (all four windows). BSA 8 nm wide; PDDA tube 0.8 nm. Green shade = mobility.



PDDA, Mw 1.0–1.5 MDa: how many BSA one chain glues

Repeat unit 161.7 g/mol, 0.50 nm, +1 → 6,185–9,278 charges; contour 3.1–4.6 µm
Coil (lp = 3 nm, ideal in the dense phase): Rg = 56–68 nm
Charge-matched (BSA net −13 to −18): 344–714 BSA per chain (≈ 499 at 1.25 MDa)
= 6.5–9.0 nm PDDA per BSA ≈ one BSA diameter: each glued BSA is wrapped
Implied PDDA 8–11 mg/mL; mesh 6.9–8.1 nm (SANS ξ ≈ 9, d = 7.7 nm)
Bulk: one coil volume holds 1,698–3,120 BSA → 4–5 chains interpenetrate

The four tight layers at hb = 20 nm

7.7-nm pitch → 1.9 mg/m² per layer (≈ a protein monolayer); 5 nm per layer
One chain neutralises all 4 layers over 81–116 nm ≈ its coil width (91–111 nm)
It crosses the 20-nm film ~46–70 times: the anchored layers are stitched into one sheet
Dry 5.7 nm; φdry 0.20 (bulk) → 0.28–0.29 (hb) → 0.44–0.53 (0.86 MPa)
→ below hb only water and ions can leave: the film densifies

Drawing rules

PDDA routed BSA to BSA: wraps each one it glues, then a tangent bridge that threads between two BSA or spans the gap (lp ≈ 3 nm)
No direction bias in the routing → isotropic where BSA are random
No chain ends in view: they leave at a window edge or go under a BSA
Adsorbed row flattened on the oil; glued row in its hollows, +1.1 nm
Same anchored molecules in every state; (3) = (2) squeezed to 0.65

What was drawn

PDDA per BSA in view: 9.2 nm (zoom), 8.9 (rest), 8.8 (hb)
Order S = <cos 2θ>: mobile +0.06 (zoom), -0.05 (rest) → isotropic anchored +0.59, film at hb +0.45 → parallel to the oil (confined)
Mobile BSA area fraction 0.39 (zoom), 0.43 (rest); φeff 0.41–0.47
5 PDDA crossings of the midplane at hb (49-nm window)